

SIMATS ENGINEERING

ASSESSMENT TOOL 1: DEBUGGING & OPTIMIZATION

COURSE NAME: Programming in Java

COURSE CODE: CSA0915- SLOT B

NAME: Lochana Sri R S

REG NO: 192521348

1. The following Java program throws an `ArithmeticException` while dividing two numbers. Identify the issue and modify the program to handle it properly.

```
public class Q1
{
    public static void main(String args[]) {
        int a = 10, b = 0;
        System.out.println(a / b);
    }
}
```

OUTPUT:

SIMATS Online Programming Lab
JAVA PROGRAMMING

```
int a = 10, b = 0;
System.out.println(a / b);
}
} Task:1. Identify the exception.2. Handle the exception using try-catch.3. Display an appropriate error message
```

Your Code

```
1 public class Q1
2 {
3     public static void main(String[] args)
4     {
5         int a = 10, b = 5;
6
7         try
8         {
9
10            System.out.println(a / b);
11        }
12        catch (ArithmeticException e)
13        {
14
15            System.out.println("Error: Division by zero is not allowed.");
16        }
17    }
18 }
19
```

Run

Save

2. The following Java program throws an `ArrayIndexOutOfBoundsException`. Identify the mistake in the loop condition and correct the program.

```
public class Q2 {  
    public static void main(String args[]) {  
        int arr[] = {10,20,30};  
        for (int i=0;i<=arr.length;i++)  
            System.out.println(arr[i]);  
    }  
}
```

OUTPUT :

SIMATS Online Programming Lab

JAVA PROGRAMMING

```
for(int i=0;i<=arr.length;i++)  
System.out.println(arr[i]);  
}  
} Task:1. Identify the error.2. Correct the loop condition.3. Display all array elements successfully.
```

Your Code

```
1 public class Q2  
2 {  
3     public static void main(String args[])  
4     {  
5         int arr[] = {10, 20, 30};  
6  
7         for (int i = 0; i < arr.length; i++)  
8         {  
9             System.out.println(arr[i]);  
10        }  
11    }  
12 }  
13
```

3. The following Java program throws a `NullPointerException`. Identify the issue and modify the program to avoid the exception.

```
public class Q3 {  
    public static void main(String args[]) {  
        String name = null;  
        System.out.println(name.length());  
    }  
}
```

```
}  
}
```

OUTPUT:

SIMATS Online Programming Lab

JAVA PROGRAMMING

Your Code

```
1 public class Q3
2 {
3     public static void main(String args[])
4     {
5         String name = null;
6
7         if (name != null)
8         {
9             System.out.println(name.length());
10        } else
11        {
12            System.out.println("Error: The string is null and has no length.");
13        }
14    }
15 }
16
```

 Run  Save

4. The following Java program throws a `NumberFormatException` while converting a string into an integer. Identify the issue and correct the program.

```
public class Q4 {  
    public static void main(String args[]) {  
        String str="12A";  
        int num=Integer.parseInt(str);  
    }  
}
```

```

System.out.println(num);
}
}

```

OUTPUT:



The screenshot shows the SIMATS Online Programming Lab interface. At the top, there's a blue header with the text "SIMATS Online Programming Lab" and a sub-header "JAVA PROGRAMMING". Below the header, there's a section titled "Your Code" containing a Java code editor. The code in the editor is as follows:

```

1 public class Q4
2 {
3     public static void main(String args[])
4     {
5         String str = "12A";
6
7         try
8         {
9             int num = Integer.parseInt(str);
10            System.out.println(num);
11        }
12        catch (NumberFormatException e)
13        {
14            System.out.println("Error: The string contains invalid non-numeric ch
15        }
16    }
17 }
18

```

Below the code editor, there are two buttons: "Run" and "Save".

5. The following Java program is intended to execute a new thread. However, it executes like a normal method. Identify the mistake and modify the program

```

class Demo extends Thread{
    public void run(){
        System.out.println("Running");
    }
}

public class Q11{
    public static void main(String args[]){
        Demo d=new Demo();
        d.run();
    }
}

```

OUTPUT:

SIMATS Online Programming Lab

JAVA PROGRAMMING

Task:1. Identify the mistake.2. Execute the thread correctly.3. Explain the difference between run() and start().

Your Code

```
1 class Demo extends Thread {
2     public void run() {
3         System.out.println("Running");
4     }
5
6     public static void main(String args[]) {
7         Demo d = new Demo();
8         d.start();
9     }
10 }
11
```

6. The following Java program implements the Runnable interface but the thread never starts. Identify the issue and correct the program.

```
class Demo implements Runnable{
    public void run(){
        System.out.println("Thread");
    }
}

public class Q12{
    public static void main(String args[]){
        Demo d=new Demo();
    }
}
```

OUTPUT:

SIMATS Online Programming Lab

JAVA PROGRAMMING

Task:1. Identify the mistake.2. Create a Thread object using the Runnable instance.3. Start the thread correctly.

Your Code

```
1 public class Demo implements Runnable
2 {
3     public void run()
4     {
5         System.out.println("Thread");
6     }
7
8     public static void main(String args[])
9     {
10        Demo d = new Demo();
11        Thread t = new Thread(d);
12        t.start();
13    }
14 }
15
16
```

 Run  Save

7. The following Java program uses try-catch but does not demonstrate the use of the finally block. Modify the program to include a finally block.

```
public class Q5 {
    public static void main(String args[]){
        try{
            int x=10/0;
        }
        catch(Exception e){
            System.out.println("Error");
        }
    }
}
```

OUTPUT:

Your Code

```
1 public class Demo
2 {
3     public static void main(String args[])
4     {
5         try
6         {
7             int x = 10 / 0;
8         }
9         catch (Exception e)
10        {
11            System.out.println("Error");
12        }
13        finally
14        {
15            System.out.println("Finally block executed");
16        }
17    }
18 }
19
```

Run

Save

Input

stdin input

Output

Error
Finally block executed

Visible Test Cases

Test Case 1

Test Case 2

Test Case 3

8. The following Java program uses `Thread.sleep()` but fails to compile because the checked exception is not handled. Modify the program to handle the exception correctly

```
public class Q8 {
    public static void main(String args[])
    {
```

```
        Thread.sleep(2000);
```

```
        System.out.println("Completed");
```

```
    }
```

```
}
```

OUTPUT:

Your Code

```
1 public class Demo
2 {
3     public static void main(String args[])
4     {
5         try
6         {
7             Thread.sleep(2000);
8         }
9         catch (InterruptedException e)
10        {
11            System.out.println("Thread was interrupted");
12        }
13        System.out.println("Completed");
14    }
15 }
16
```

Run

Save

Input

stdin input

Output

Completed

Visible Test Cases

Test Case 1

Test Case 2

Test Case 3

9. The following program creates two thread classes but does not execute them. Identify the mistake and modify the program.

```
class A extends Thread
{
    public void run() {
        System.out.println("Thread A is Running");
    }
}

class B extends Thread {
    public void run() {
        System.out.println("Thread B is Running");
    }
}

public class Q9
{
    public static void main(String args[]) {
        A a = new A();
        B b = new B();
    }
}
```

OUTPUT:

SIMATS Online Programming Lab

JAVA PROGRAMMING

Your Code

```
1 class A extends Thread
2 {
3     public void run()
4     {
5         System.out.println("Thread A is Running");
6     }
7
8     public static void main(String args[])
9     {
10        A a = new A();
11        B b = new B();
12
13        a.start();
14        b.start();
15    }
16 }
17
18 class B extends Thread
19 {
20     public void run()
21     {
22         System.out.println("Thread B is Running");
23     }
24 }
25
```

▶ Run

📁 Save

Input

stdin input

Output

Thread A is Running
Thread B is Running

Visible Test Cases

Test Case 1

Test Case 2

Test Case 3

10. The following Java program allows two threads to update the same bank account balance simultaneously, resulting in inconsistent output. Identify the synchronization issue and modify the program using the synchronized keyword.

```
class Account {
    int balance = 1000;
    void withdraw(int amount) {
        if(balance >= amount) {
            balance = balance - amount;
            System.out.println("Remaining Balance : " + balance);
        }
        else {
            System.out.println("Insufficient Balance");
        }
    }
}
class Customer extends Thread {
    Account acc;
    Customer(Account acc) {
        this.acc = acc;
    }
    public void run() {
        acc.withdraw(700);
    }
}
public class Q10 {
    public static void main(String args[]) {
        Account obj = new Account();
        Customer c1 = new Customer(obj);
        Customer c2 = new Customer(obj);
        c1.start();
        c2.start();
    }
}
OUTPUT:
```

SIMATS Online Programming Lab

JAVA PROGRAMMING

Your Code

```
1 public class Account
2 {
3     int balance = 1000;
4
5     synchronized void withdraw(int amount)
6     {
7         if (balance >= amount)
8         {
9             balance = balance - amount;
10            System.out.println("Remaining Balance : " + balance);
11        }
12        else
13        {
14            System.out.println("Insufficient Balance");
15        }
16    }
17
18    public static void main(String args[])
19    {
20        Account obj = new Account();
21        Customer c1 = new Customer(obj);
22        Customer c2 = new Customer(obj);
23
24        c1.start();
25        c2.start();
26    }
27 }
28
29 class Customer extends Thread
30 {
31     Account acc;
```

Input

stdin input

Output

```
Remaining Balance : 300
Insufficient Balance
```

Visible Test Cases

Test Case 1

Test Case 2

Test Case 3